

MAST32001 Computational Statistics - Lecture 5

Permutation principles and randomization tests

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Permutation principles

Today

The central question is:

When is it legitimate to rearrange the observed data?

We will study

- ▶ independence and exchangeability;
- ▶ the permutation principle;
- ▶ exact, Monte Carlo, and asymptotic tests;
- ▶ two-sample location tests;
- ▶ rank-based tests;
- ▶ tests for k groups;
- ▶ permutation tests of independence.

The key idea is simple:

Under the null hypothesis, some labels or observations may be interchangeable.

The mathematical work is to identify **exactly which transformations are valid**.

A generic testing problem

Suppose we observe data X and want to test

$$H_0 \quad \text{versus} \quad H_1.$$

Choose a statistic

$$T = T(X)$$

such that large values provide evidence against H_0 .

A classical test requires the null distribution

$$\mathcal{L}_0(T).$$

Sometimes this distribution is known analytically.

Sometimes it is only approximately known.

Permutation tests construct a null distribution by applying transformations that are valid under H_0 .

Three notions of a null distribution

We will distinguish three ideas.

Exact finite-sample distribution

$$P_{H_0}(T \geq t)$$

is known or generated correctly for the actual sample size.

Asymptotic distribution

$$a_n\{T - b_n\} \xrightarrow{d} G$$

as $n \rightarrow \infty$.

Monte Carlo approximation We simulate from a finite-sample reference distribution and approximate its probabilities numerically.

These concepts should not be confused:

exactness of the test $\neq$ exact enumeration of all permutations.

Independence and exchangeability

Independence

Random variables $X_1, \dots, X_n$ are **independent** if for all measurable sets $A_1, \dots, A_n$,

$$P(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^n P(X_i \in A_i).$$

If they also have the same distribution F , they are **independent and identically distributed**:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F.$$

Independence describes the factorization of the joint distribution.

It is not, by itself, the property used directly by permutation tests.

Exchangeability

A random vector

$$(X_1, \dots, X_n)$$

is **exchangeable** if for every permutation π of $\{1, \dots, n\}$,

$$(X_1, \dots, X_n) \stackrel{d}{=} (X_{\pi(1)}, \dots, X_{\pi(n)}).$$

Thus the joint distribution does not change when the coordinates are relabelled.

Exchangeability is a **symmetry property of the joint distribution**.

IID implies exchangeability

If

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F,$$

then

$$P(X_1 \in dx_1, \dots, X_n \in dx_n) = \prod_{i=1}^n F(dx_i).$$

For every permutation π ,

$$\prod_{i=1}^n F(dx_i) = \prod_{i=1}^n F(dx_{\pi(i)}).$$

Hence

$$(X_1, \dots, X_n) \stackrel{d}{=} (X_{\pi(1)}, \dots, X_{\pi(n)}).$$

Therefore

iid $\implies$ exchangeable.

The converse is false.

Exchangeable but dependent

Let

$$Z \sim N(0, 1), \quad \varepsilon_1, \dots, \varepsilon_n \stackrel{\text{iid}}{\sim} N(0, 1),$$

independently, and define

$$X_i = Z + \varepsilon_i.$$

Then all X_i have the same marginal distribution and the joint distribution is unchanged by permutations.

So $X_1, \dots, X_n$ are exchangeable.

But

$$\text{Cov}(X_i, X_j) = \text{Var}(Z) = 1, \quad i \neq j.$$

Thus exchangeability does **not** imply independence.

Why exchangeability matters

Suppose G is a collection of transformations of the data.

For a permutation test we want, under H_0 ,

$$gX \stackrel{d}{=} X \quad \text{for every } g \in G.$$

Then the observed dataset and its transformed versions are equivalent under the null.

This is the general **invariance principle** behind permutation tests.

The relevant symmetry may concern

- ▶ observations;
- ▶ group labels;
- ▶ one variable relative to another;
- ▶ signs;
- ▶ residuals.

Not every null hypothesis implies every kind of permutation invariance.

Important warning

A permutation test is not justified merely because we can call

```
sample()
```

on something.

We must first ask:

What is exchangeable under H_0 ?

For example:

- ▶ iid observations can be reordered;
- ▶ under a two-sample equal-distribution null, group labels can be reassigned;
- ▶ under independence, one variable can be permuted relative to the other.

The permutation scheme is part of the statistical model.

The permutation principle

A group of transformations

Let

$$G = \{g_1, \dots, g_M\}$$

be a finite collection of transformations, usually permutations.

Assume that under H_0 ,

$$gX \stackrel{d}{=} X \quad \forall g \in G.$$

For each transformed dataset calculate

$$T(gX).$$

The collection

$$\{T(gX) : g \in G\}$$

is the **permutation distribution** of the statistic conditional on the observed orbit.

Permutation p-value

Suppose large T is evidence against H_0 . If all M transformations are enumerated, define

$$p_{\text{perm}} = \frac{1}{M} \sum_{g \in G} I\{T(gX) \geq T(X)\}.$$

For a two-sided test, the statistic itself should encode the relevant two-sided departure, for example

$$T = |\bar{X} - \bar{Y}|.$$

The test rejects for sufficiently small p_{perm} .

Why can the test be exact?

Under H_0 , the transformed datasets

$$\{gX : g \in G\}$$

are equally plausible members of the orbit generated by the observed data.

Consequently, the rank of $T(X)$ among

$$\{T(gX) : g \in G\}$$

has the correct null behaviour.

This yields finite-sample type-I error control, subject to issues such as discreteness and ties.

The argument comes from **symmetry under the null**, not from a large-sample approximation.

Exact does not mean continuous

Permutation distributions are discrete.

Therefore the set of attainable p-values is discrete as well.

If there are M distinct permutations, a p-value may only take values on a finite grid.

Consequently, a level- α permutation test may satisfy

$$P_{H_0}(\text{reject}) \leq \alpha$$

rather than exactly α .

This conservativeness is a consequence of discreteness, not a failure of the permutation principle.

Enumeration versus random permutations

Often

$$M = |G|$$

is too large to enumerate.

Instead sample transformations

$$g_1, \dots, g_B$$

at random and compute

$$T_b = T(g_b X).$$

A standard Monte Carlo p-value is

$$\hat{p} = \frac{1 + \sum_{b=1}^B I(T_b \geq T_{\text{obs}})}{B + 1}$$

when the observed arrangement is included as one of the $B + 1$ possibilities.

The +1 correction prevents a simulated p-value of zero.

Monte Carlo uncertainty

Conditional on the data, the number of sampled permutation statistics exceeding T_{obs} is approximately binomial.

Thus, roughly,

$$\text{SE}(\hat{p}) \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{B}}.$$

If $\hat{p} \approx 0.05$,

$$B = 999$$

gives a Monte Carlo standard error of about 0.007.

With

$$B = 9999,$$

it is about 0.002.

The statistical test may be exact in principle while its numerically estimated p-value still has Monte Carlo error.

TASK: what may be permuted?

For each situation, decide what is exchangeable under the null.

1. Two independent groups with identical distributions under H_0 .
2. Paired before/after measurements under a no-effect symmetry assumption.
3. Two variables X and Y , testing $X \perp Y$.
4. A time series with strong serial dependence.

For each case answer:

- ▶ What would you permute?
- ▶ What must remain fixed?
- ▶ Is unrestricted permutation plausible?

Discuss with your neighbour.

TASK: discussion

Possible answers:

1. **Two groups:** pool observations and permute group labels, keeping group sizes fixed.
2. **Paired data:** do not freely permute individual observations; paired structure must be preserved. Depending on assumptions, signs or within-pair labels may be randomized.
3. **Independence:** keep the X 's fixed and permute the Y 's relative to them.
4. **Time series:** unrestricted permutation destroys serial dependence and is generally invalid.

The lesson:

Permutation validity depends on the null symmetry and the data structure.

Two-sample location problems

Two independent samples

Suppose

$$X_1, \dots, X_{n_1} \sim F, \quad Y_1, \dots, Y_{n_2} \sim G,$$

with the two samples independent.

A common scientific question is whether the groups differ in location.

Possible null hypotheses include

$$H_0 : \mu_X = \mu_Y$$

or the stronger hypothesis

$$H_0 : F = G.$$

These are not the same null hypothesis.

This distinction is crucial for permutation testing.

Equality of distributions gives label exchangeability

Under

$$H_0 : F = G,$$

the pooled observations

$$Z = (X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2})$$

are iid from the same distribution.

Therefore the group labels are exchangeable.

There are

$$\binom{n_1 + n_2}{n_1}$$

possible assignments of n_1 observations to group 1.

A permutation test may compare the observed statistic with its values under these assignments.

Difference in means

A natural statistic is

$$T = \bar{X} - \bar{Y}.$$

For a two-sided test we may use

$$|T| = |\bar{X} - \bar{Y}|.$$

Permutation algorithm:

1. Pool all observations.
2. Randomly choose n_1 observations for group 1.
3. Assign the remaining n_2 to group 2.
4. Recompute T .
5. Repeat.

Under $F = G$, this gives a finite-sample permutation test.

R: two-sample permutation test

```
set.seed(20260812)
x <- c(4.2, 5.1, 5.5, 5.8, 6.2)
y <- c(5.7, 6.0, 6.4, 6.8, 7.1)
t_obs <- abs(mean(x) - mean(y))
z <- c(x, y)
n1 <- length(x)
B <- 9999

t_perm <- replicate(B, {
  ind <- sample.int(length(z), n1, replace = FALSE)
  abs(mean(z[ind]) - mean(z[-ind]))
})

p_perm <- (1 + sum(t_perm >= t_obs)) / (B + 1)
c(statistic = t_obs, p_value = p_perm)

## statistic    p_value
##      1.0400      0.0469
```

What exactly is being tested?

The simple label permutation argument is exact under

$$H_0 : F = G.$$

But suppose we only assume

$$H_0 : E(X) = E(Y)$$

while allowing

$$F \neq G.$$

Then the observations are generally **not exchangeable across groups**.

For example, the groups could have equal means but different variances.

Thus a naive permutation test of the raw mean difference need not be finite-sample exact for the weaker equal-means null.

This distinction between the **scientific null** and the **exchangeability null** is important.

Studentization

One way to obtain better robustness for a location comparison is to use a studentized statistic such as

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{s_X^2/n_1 + s_Y^2/n_2}}.$$

The permutation distribution of this statistic is still exactly justified by complete label exchangeability under $F = G$.

Under weaker conditions, studentized permutation tests can also have useful **asymptotic** validity.

This is a good example of the distinction

finite-sample exactness versus asymptotic validity.

Rank tests

Why ranks?

Suppose the pooled observations are

$$Z_1, \dots, Z_N, \quad N = n_1 + n_2.$$

Replace each observation by its rank among all N observations:

$$R_i = \text{rank}(Z_i).$$

For continuous data there are no ties with probability one, and the ranks are simply

$$1, \dots, N.$$

Ranks discard the numerical spacing between observations and retain only their ordering.

They are therefore invariant under strictly increasing transformations.

A small rank example

Suppose

$$x = (2.1, 4.8, 5.0), \quad y = (3.2, 5.7, 7.1).$$

The pooled ordered observations are

$$2.1, 3.2, 4.8, 5.0, 5.7, 7.1.$$

Hence the ranks of the x -observations are

$$1, 3, 4.$$

Their rank sum is

$$W = 1 + 3 + 4 = 8.$$

If the x -group tends to contain smaller observations, W tends to be small.

If it tends to contain larger observations, W tends to be large.

Wilcoxon rank-sum statistic

Let R_i be the pooled rank of observation X_i .

The **Wilcoxon rank-sum statistic** is

$$W = \sum_{i=1}^{n_1} R_i.$$

Under the equal-distribution null and continuous observations, every choice of n_1 ranks from

$$1, \dots, N$$

is equally likely.

Therefore the null distribution of W is a permutation distribution.

Its expectation is

$$E_0(W) = \frac{n_1(N+1)}{2}.$$

Variance of the rank sum

Without ties,

$$\text{Var}_0(W) = \frac{n_1 n_2 (N + 1)}{12}.$$

Therefore an asymptotic standardized statistic is

$$Z = \frac{W - \frac{n_1(N+1)}{2}}{\sqrt{n_1 n_2 (N + 1)/12}}.$$

For sufficiently large samples,

$$Z \dot{\sim} N(0, 1)$$

under H_0 .

Thus the same rank statistic can be used with

- ▶ an exact permutation distribution;
- ▶ a Monte Carlo permutation distribution;
- ▶ an asymptotic normal approximation.

Rank-sum and Mann–Whitney

An equivalent statistic is

$$U = W - \frac{n_1(n_1 + 1)}{2}.$$

It counts, up to conventions for ties, pairwise comparisons between the two groups.

For continuous distributions,

$$U = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} I(X_i > Y_j)$$

for one orientation of the statistic.

Hence

$$\frac{U}{n_1 n_2}$$

estimates a probability of stochastic ordering such as

$$P(X > Y).$$

The Wilcoxon rank-sum and Mann–Whitney tests are therefore two views of essentially the same test.

What does the rank-sum test test?

Its exact permutation null is

$$H_0 : F = G.$$

It is often described as a test for a location shift, but that interpretation requires additional structure.

For example, under a shift model

$$G(x) = F(x - \Delta),$$

testing

$$\Delta = 0$$

is equivalent to testing equality of the distributions.

Without such a model, rejection can be caused by more general distributional differences.

A rank test is not automatically a test only of medians.

Ties

If observations are tied, assign average ranks.

Example:

2, 4, 4, 7

has ranks

1, 2.5, 2.5, 4.

Ties affect the exact distribution and the variance formula.

Software therefore needs to account for them.

Conceptually, however, the permutation principle remains:

Under the null, which group assignments are legitimate for the observed values?

R: rank-sum test

```
x <- c(2.1, 4.8, 5.0)
y <- c(3.2, 5.7, 7.1)
wilcox.test(x, y, exact = TRUE)
```

```
##
## Wilcoxon rank sum exact test
##
## data:  x and y
## W = 2, p-value = 0.4
## alternative hypothesis: true location shift is not equal to 0
```

We can also construct the statistic ourselves:

```
z <- c(x, y)
r <- rank(z)
W <- sum(r[seq_along(x)])
W
```

```
## [1] 8
```

The software implementation and the permutation principle are two descriptions of the same underlying combinatorial idea.

Exact, asymptotic, and permutation inference

Three routes for the same statistic

Consider a statistic T_n .

Exact analytic test

Derive its finite-sample null distribution mathematically.

Asymptotic test

Find a limiting distribution:

$$T_n \Rightarrow T.$$

Permutation test

Generate the finite-sample reference distribution using transformations justified by the null.

These are different ways of calibrating a statistic.

The statistic itself does not determine whether the resulting test is exact or asymptotic.

Example: rank sum

For the Wilcoxon rank-sum statistic W :

1. Its exact distribution can be obtained combinatorially.
2. It can be generated by permuting group labels.
3. For large samples, it can be approximated by a normal distribution.

Thus

same statistic + different reference distributions.

This is a useful general principle in computational statistics.

What does “exact” mean?

An **exact test** controls type-I error at the desired level in finite samples under its stated assumptions.

It does **not** mean:

- ▶ the p-value has no numerical error;
- ▶ all permutations must be enumerated;
- ▶ no discreteness is present;
- ▶ the null hypothesis is scientifically appropriate.

Always ask:

Exact under which assumptions?

More than two groups

The k -sample problem

Suppose there are k independent groups:

$$X_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, n_i.$$

A strong null hypothesis is

$$H_0 : F_1 = \dots = F_k.$$

Under this null, all

$$N = \sum_{i=1}^k n_i$$

observations are exchangeable with respect to group labels.

A permutation must preserve the group sizes

$$n_1, \dots, n_k.$$

ANOVA-type statistic

One possible statistic measures between-group variation:

$$T = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X})^2.$$

Large values indicate that the group means are separated.

A permutation test can be constructed by repeatedly assigning the pooled observations to groups of sizes

$$n_1, \dots, n_k$$

and recalculating T .

Under equality of the full distributions, the resulting label permutation test is finite-sample exact.

Rank-based k -sample testing

For robustness to scale and outliers, replace the observations by pooled ranks.

Let

$$R_{ij}$$

be the rank of observation X_{ij} among all N observations.

Let

$$\bar{R}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} R_{ij}.$$

If the groups have the same distribution, their mean ranks should be similar.

This leads to the **Kruskal–Wallis statistic**.

Kruskal–Wallis statistic

Without ties,

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i \left(\bar{R}_i - \frac{N+1}{2} \right)^2.$$

Equivalently, if

$$R_i = \sum_{j=1}^{n_i} R_{ij},$$

then

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1).$$

Large H means that the rank distributions differ between groups.

Kruskal–Wallis null distributions

Under

$$H_0 : F_1 = \cdots = F_k,$$

we may obtain a finite-sample permutation distribution by permuting the group labels.

For sufficiently large samples, the usual approximation is

$$H \dot{\sim} \chi_{k-1}^2.$$

Thus again we have

permutation calibration or asymptotic calibration.
--

With ties, a correction is required.

R: k -sample rank test

```
set.seed(20260812)
g <- factor(rep(c("A", "B", "C"), each = 8))
x <- c(
  rnorm(8, 0.0, 1),
  rnorm(8, 0.7, 1),
  rnorm(8, 1.2, 1)
)
kruskal.test(x ~ g)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  x by g
## Kruskal-Wallis chi-squared = 1.28, df = 2, p-value = 0.5273
```

The usual R output uses the asymptotic chi-squared reference distribution.

A permutation version would keep x fixed and randomly reassign the group labels while preserving the group sizes.

TASK: exact or asymptotic?

Consider three groups of sizes

$$n_1 = n_2 = n_3 = 6.$$

You use the Kruskal–Wallis statistic.

Two proposed analyses are:

A. Compare H with a χ^2_2 distribution.

B. Randomly permute the group labels 9999 times and compare the observed H with the permutation values.

Discuss:

- ▶ Which analysis is asymptotic?
- ▶ Which is based on finite-sample exchangeability?
- ▶ Is analysis B an exact enumeration?
- ▶ What assumptions make B valid?
- ▶ What numerical uncertainty remains in B?

TASK: solution

A uses the asymptotic approximation

$$H \sim \chi^2_2.$$

B approximates the finite-sample permutation distribution by Monte Carlo.

It is not an enumeration of all possible assignments.

Its validity comes from exchangeability of the group labels under

$$H_0 : F_1 = F_2 = F_3.$$

Because only 9999 random permutations are used, the reported p-value has Monte Carlo uncertainty.

Thus

finite-sample permutation principle

and

Monte Carlo approximation

can occur simultaneously.

Testing independence

Independence of two random variables

Random variables X and Y are independent if

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

for all measurable sets A, B .

Equivalently, when a joint density exists,

$$f_{X,Y}(x, y) = f_X(x)f_Y(y).$$

We write

$$X \perp Y.$$

Independence is stronger than zero covariance.

Zero correlation is not independence

If X and Y have finite variances,

$$X \perp Y \implies \text{Cov}(X, Y) = 0.$$

The converse is false.

For example, let

$$X \sim N(0, 1), \quad Y = X^2.$$

Then

$$E(XY) = E(X^3) = 0,$$

so

$$\text{Cov}(X, Y) = 0.$$

But Y is completely determined by X , so X and Y are certainly not independent.

Independence and permutation

Suppose we observe iid pairs

$$(X_1, Y_1), \dots, (X_n, Y_n).$$

Under

$$H_0 : X \perp Y,$$

the joint distribution factorizes:

$$\prod_{i=1}^n f_X(x_i) f_Y(y_i).$$

For every permutation π ,

$$\prod_{i=1}^n f_X(x_i) f_Y(y_{\pi(i)})$$

is the same product.

Therefore under independence,

$$\{(X_i, Y_i)\}_{i=1}^n \quad \text{has the same distribution as} \quad \{(X_i, Y_{\pi(i)})\}_{i=1}^n.$$

This gives a natural permutation principle for testing independence.

Correlation as a test statistic

A familiar statistic is Pearson correlation:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}}.$$

For a two-sided test, use

$$T = |r|.$$

Permutation algorithm:

1. Keep $X_1, \dots, X_n$ fixed.
2. Permute $Y_1, \dots, Y_n$.
3. Recompute r .
4. Compare $|r_{\text{obs}}|$ with the permutation distribution.

R: permutation test of independence

```
set.seed(20260812)
n <- 30
x <- rnorm(n)
y <- 0.5 * x + rnorm(n)
t_obs <- abs(cor(x, y))
B <- 9999
t_perm <- replicate(B, {
  abs(cor(x, sample(y)))
})
p_perm <- (1 + sum(t_perm >= t_obs)) / (B + 1)
c(correlation = cor(x, y), p_value = p_perm)
```

```
## correlation      p_value
##      0.413225      0.021900
```

This tests independence using correlation as the chosen measure of association.

Power will be good for some alternatives and poor for others.

A rank-based measure: Spearman correlation

Let

$$R_i = \text{rank}(X_i), \quad S_i = \text{rank}(Y_i).$$

The **Spearman rank correlation** is simply the Pearson correlation of the ranks:

$$\rho_S = \text{cor}(R, S).$$

It measures monotone association rather than only linear association.

Because ranks are unchanged by strictly increasing transformations, Spearman correlation is less dependent on the numerical scales of X and Y .

Spearman's rho without ties

When there are no ties, let

$$d_i = R_i - S_i.$$

Then

$$\rho_S = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}.$$

Under independence, the pairing between the ranks of X and Y is exchangeable.

Therefore a permutation test can be obtained by permuting one rank vector relative to the other.

Again, the rank statistic and the permutation principle fit naturally together.

R: Pearson versus Spearman

```
set.seed(20260812)
x <- runif(40, -2, 2)
y <- x^3 + rnorm(40, sd = 1.5)
cor(x, y, method = "pearson")
```

```
## [1] 0.7883031
```

```
cor(x, y, method = "spearman")
```

```
## [1] 0.85197
```

R: Pearson versus Spearman

```
cor.test(x, y, method = "spearman", exact = FALSE)

##
## Spearman's rank correlation rho
##
## data:  x and y
## S = 1578, p-value = 3.136e-12
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.85197
```

Pearson correlation focuses on linear association.

Spearman correlation focuses on monotone association.

Neither statistic detects every possible form of dependence.

Independence is broader than correlation

A permutation principle does not tell us which statistic to use.

It tells us how to generate a valid null reference distribution once a statistic has been chosen.

For independence testing we could use

- ▶ Pearson correlation;
- ▶ Spearman correlation;
- ▶ Kendall-type statistics;
- ▶ distance-based measures;
- ▶ kernel-based measures.

Different statistics target different alternatives.

Null symmetry determines the permutation scheme; the statistic determines sensitivity.
--

Interactive R investigation

R TASK: same permutation principle, different statistics

Generate

```
set.seed(20260812)
n <- 60
x <- runif(n, -2, 2)
y <- x^2 + rnorm(n, sd = 0.4)
```

Work in pairs.

1. Plot y against x.
2. Compute Pearson and Spearman correlations.
3. For each statistic, perform a permutation test with $B \leftarrow 1999$.
4. What do the two tests conclude?
5. Does a large p-value establish independence?

Use

```
sample(y)
```

to break the observed pairing.

R TASK: solution – the data

```
set.seed(20260812)

n <- 60
x <- runif(n, -2, 2)
y <- x^2 + rnorm(n, sd = 0.4)
```

The data were generated from

$$Y = X^2 + \varepsilon.$$

Thus X and Y are clearly **not independent**.

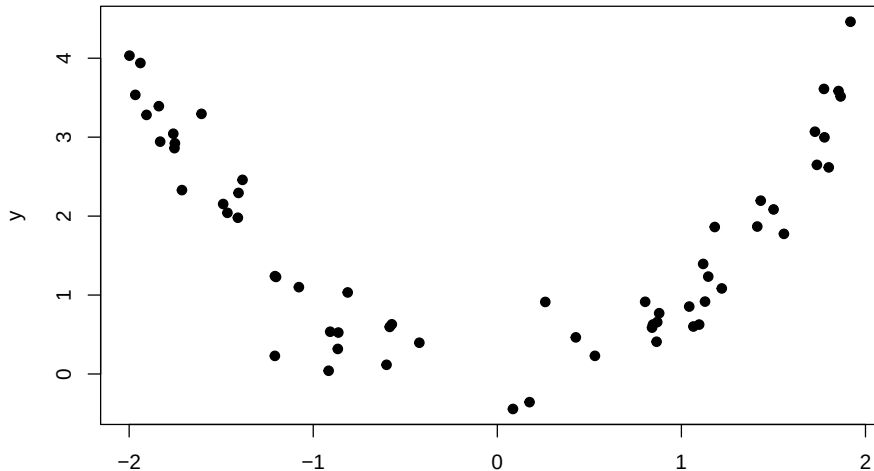
However, the dependence is not monotone:

- ▶ large negative and large positive values of X both tend to produce large Y ;
- ▶ values of X close to zero tend to produce small Y .

What should we expect from Pearson and Spearman correlation?

R TASK: solution – look at the data

Strong dependence, but little monotone association



R TASK: solution – observed statistics

We use two measures of association:

$$T_P = |\text{cor}(X, Y)| \quad \text{and} \quad T_S = |\text{cor}(\text{rank}(X), \text{rank}(Y))|.$$

In R:

```
pearson_obs <- abs(cor(x, y, method = "pearson"))
spearman_obs <- abs(cor(x, y, method = "spearman"))
c(Pearson = pearson_obs, Spearman = spearman_obs)
```

```
##   Pearson   Spearman
```

```
## 0.1063641 0.0706863
```

Both statistics measure particular forms of association.

Neither is designed specifically for a U-shaped relationship.

R TASK: solution – permutation tests

Under

$$H_0 : X \perp Y,$$

we may permute the Y -values relative to the X -values.

```
B <- 1999
pearson_perm <- replicate(
  B, abs(cor(x, sample(y), method = "pearson")))
)
spearman_perm <- replicate(
  B, abs(cor(x, sample(y), method = "spearman")))
)
p_pearson <- (1 + sum(pearson_perm >= pearson_obs)) / (B + 1)
p_spearman <- (1 + sum(spearman_perm >= spearman_obs)) / (B + 1)
c(Pearson = p_pearson, Spearman = p_spearman)
```

```
## Pearson Spearman
```

```
## 0.434 0.583
```

The permutation scheme is valid under independence.

Connecting the examples

Three permutation schemes

Two groups

Under equality of distributions:

permute group labels.

k groups

Under equality of all group distributions:

permute group labels while preserving group sizes.

Independence

Under

$$X \perp Y :$$

permute one variable relative to the other.

The implementation looks similar, but the underlying null symmetry is different.

Rank tests through the permutation lens

Rank tests are sometimes presented as a separate collection of named procedures.

A more unified view is:

1. transform observations to ranks;
2. choose a statistic based on those ranks;
3. determine its null distribution.

Examples:

two groups	→	Wilcoxon rank sum,
k groups	→	Kruskal–Wallis,
association	→	Spearman correlation.

Permutation ideas explain naturally why these statistics have distribution-free null behaviour under suitable assumptions.

Distribution-free does not mean assumption-free

A rank test may be called **distribution-free** because its null distribution does not depend on an unknown continuous distribution F .

But assumptions remain.

Examples include

- ▶ independent observations;
- ▶ exchangeability under the null;
- ▶ continuity when using no-tie formulas;
- ▶ the appropriate sampling design.

Thus

distribution-free $\neq$ assumption-free.

Exact versus asymptotic: summary table

Method	Reference distribution	Main justification
Permutation	finite-sample rearrangements	null invariance
Monte Carlo permutation	sampled rearrangements	null invariance + simulation
Asymptotic	limiting law	$n \rightarrow \infty$

An exact permutation principle can be implemented with a Monte Carlo approximation.

An asymptotic test may use the same statistic but a different reference distribution.

Practical issues

How many permutations?

If all permutations can be enumerated, enumeration removes Monte Carlo error.

Otherwise choose B according to the resolution needed.

Since

$$\hat{p} = \frac{1 + K}{B + 1},$$

the smallest attainable Monte Carlo p-value is

$$\frac{1}{B + 1}.$$

For example,

$$B = 999 \quad \Rightarrow \quad p_{\min} = 0.001.$$

If very small p-values matter, substantially more permutations are needed.

Reproducibility

Permutation tests use random number generation when permutations are sampled.

Therefore use a seed:

```
set.seed(20260812)
```

and report

- ▶ the statistic;
- ▶ the permutation scheme;
- ▶ the number of random permutations B ;
- ▶ the p-value formula;
- ▶ any handling of ties or restricted permutations.

A permutation test is an algorithm and should be reproducible like any other computational analysis.

Common mistakes

1. Permuting observations without identifying the null symmetry.
2. Treating exchangeability as synonymous with independence.
3. Claiming that a test of $F = G$ is automatically a test only of equal means or medians.
4. Ignoring paired, clustered, spatial, or temporal structure.
5. Reporting a Monte Carlo p-value of zero.
6. Forgetting Monte Carlo uncertainty.
7. Assuming a rank test is assumption-free.
8. Using a statistic that has little power for the alternative of interest.

Final synthesis

A general recipe

For a permutation test:

1. State the null hypothesis precisely.
2. Identify the transformations that leave the null distribution invariant.
3. Choose a statistic sensitive to relevant alternatives.
4. Calculate the observed statistic.
5. Generate its permutation distribution.
6. Compare the observed value with that distribution.
7. Quantify Monte Carlo uncertainty if permutations are sampled.
8. Interpret the conclusion relative to the actual null hypothesis.

The most important step is usually **Step 2**.

Take-home messages

- ▶ Independence and exchangeability are different concepts.
- ▶ IID observations are exchangeable, but exchangeability does not imply independence.
- ▶ Permutation tests are based on invariance under the null.
- ▶ Finite-sample exactness and asymptotic validity are different notions.
- ▶ Monte Carlo permutation tests approximate an exact permutation distribution numerically.
- ▶ Rank tests fit naturally into the permutation framework.
- ▶ Wilcoxon rank sum handles two groups; Kruskal–Wallis extends the rank idea to k groups.
- ▶ Independence can be tested by permuting one variable relative to another.
- ▶ A valid permutation scheme does not rescue a poorly chosen test statistic.

Next perspective

Permutation tests exploit **symmetry** to construct reference distributions.

This idea will reappear throughout computational statistics:

identify structure $\longrightarrow$ simulate under the null $\longrightarrow$ compare with what was observed.